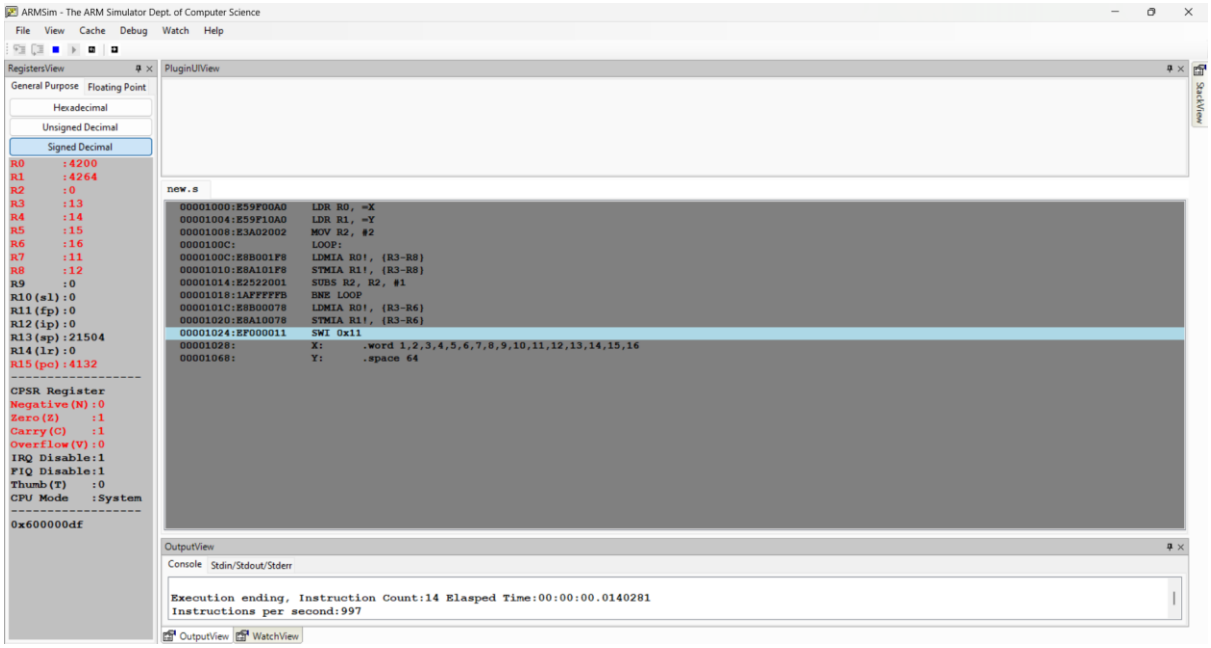
Department of Computer Science & Engineering
Microprocessor & Computer Architecture Lab

Lab 5 Programs
UE24CS251B

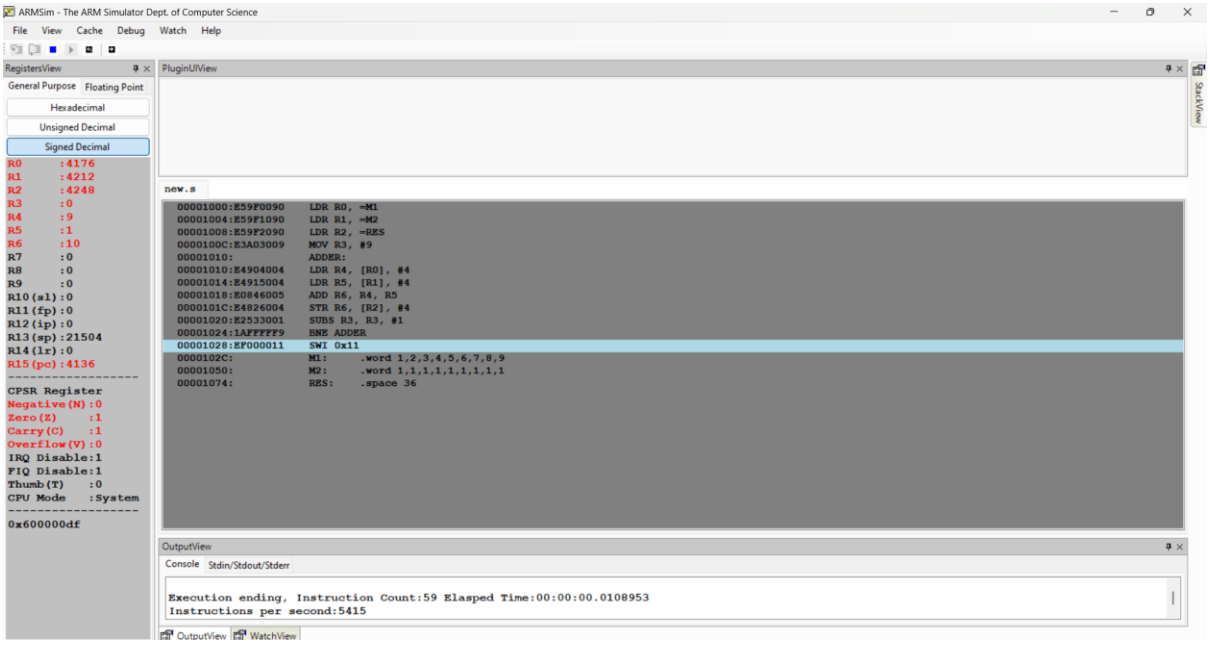
Q1. Write a program in ARM7TDMI-ISA to transfer a block of 16 words stored at memory location X to memory location Y using Load Multiple and Store Multiple instructions. The rate of transfer is 24 bytes

```
LDR R0, =X
LDR R1, =Y
MOV R2, #2
LOOP:
LDMIA R0!, {R3-R8}
STMIA R1!, {R3-R8}
SUBS R2, R2, #1
BNE LOOP
LDMIA R0!, {R3-R6}
STMIA R1!, {R3-R6}
SWI 0x11
X: .word 1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16
Y: .space 64
```



Q2. Write a program to perform 3X3 matrix addition.

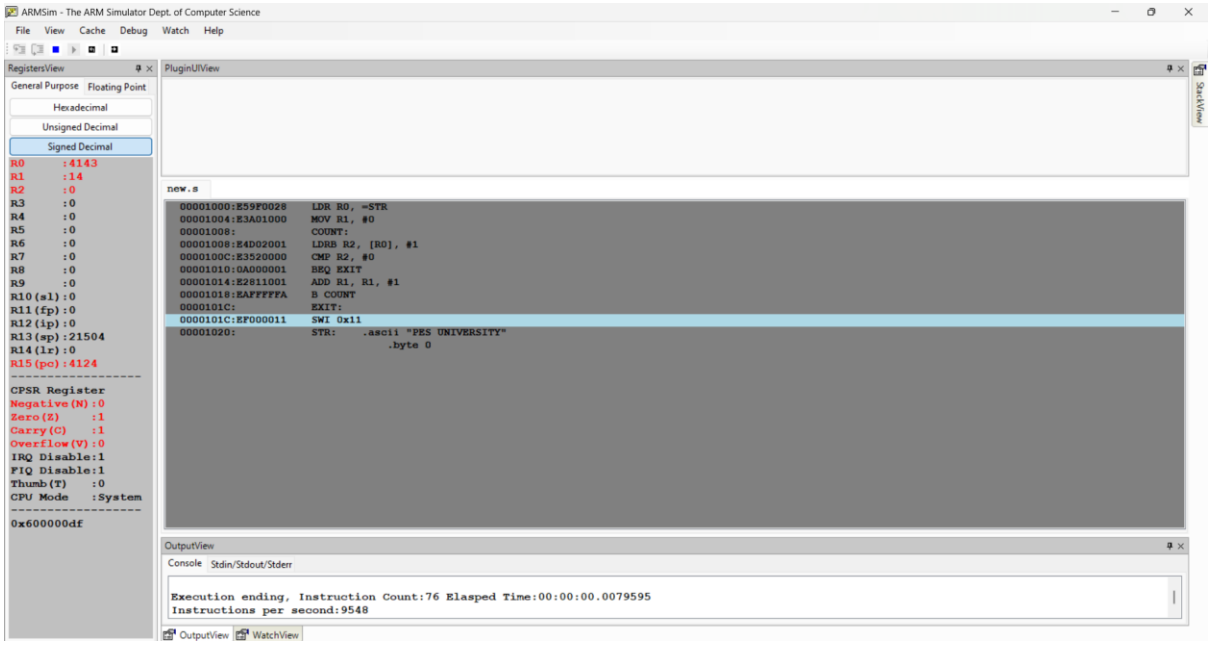
```
LDR R0, =M1
LDR R1, =M2
LDR R2, =RES
MOV R3, #9
ADDER:
LDR R4, [R0], #4
LDR R5, [R1], #4
ADD R6, R4, R5
STR R6, [R2], #4
SUBS R3, R3, #1
BNE ADDER
SWI 0x11
M1: .word 1,2,3,4,5,6,7,8,9
M2: .word 1,1,1,1,1,1,1,1,1
RES: .space 36
```



Q3. Write a program to find the length of a string.

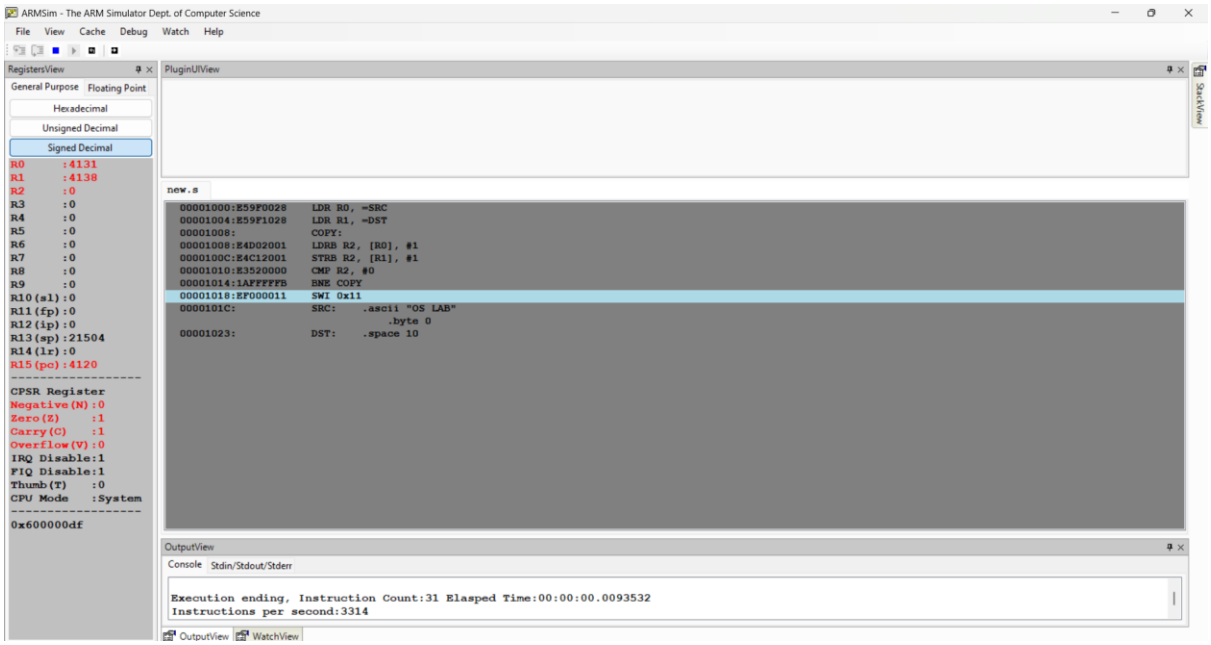
```
LDR R0, =STR
MOV R1, #0
COUNT:
LDRB R2, [R0], #1
CMP R2, #0
BEQ EXIT
ADD R1, R1, #1
B COUNT
EXIT:
```

```
SWI 0x11
STR: .ascii "PES UNIVERSITY"
        .byte 0
```



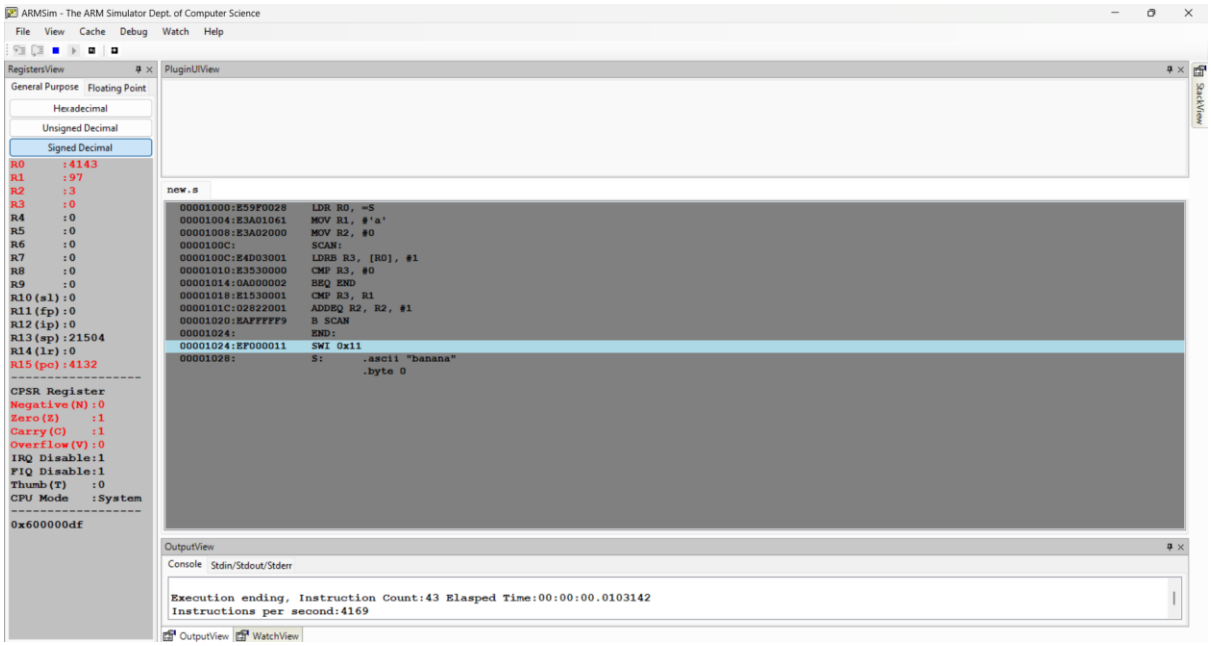
Q4. Write a program to copy a string from one location to another location.

```
LDR R0, =SRC
LDR R1, =DST
COPY:
LDRB R2, [R0], #1
STRB R2, [R1], #1
CMP R2, #0
BNE COPY
SWI 0x11
SRC: .ascii "OS LAB"
        .byte 0
DST: .space 10
```



Q5. Write an ALP to find how many times the given character is present in a string.

```
LDR R0, =S
MOV R1, #'a'
MOV R2, #0
SCAN:
LDRB R3, [R0], #1
CMP R3, #0
BEQ END
CMP R3, R1
ADDEQ R2, R2, #1
B SCAN
END:
SWI 0x11
S: .ascii "banana"
    .byte 0
```


ARMSim - The ARM Simulator Dept. of Computer Science
File View Cache Debug Watch Help
RegistersView
General Purpose Floating Point
Hexadecimal
Unsigned Decimal
Signed Decimal
R0 :4143
R1 :97
R2 :3
R3 :0
R4 :0
R5 :0
R6 :0
R7 :0
R8 :0
R9 :0
R10(sl):0
R11(fp):0
R12(ip):0
R13(sp):21504
R14(lr):0
R15(pc):4132
CPSR Register
Negative(N):0
Zero(Z) :1
Carry(C) :1
Overflow(V):0
IRQ Disable:1
FIQ Disable:1
Thumb(T) :0
CPU Mode :System
0x600000df
PluginUIView
StackView
new.s
00001000:E59F0028 LDR R0, =S
00001004:E3A01061 MOV R1, #'a'
00001008:E3A02000 MOV R2, #0
0000100C: SCAN:
0000100C:E4D03001 LDRB R3, [R0], #1
00001010:E3530000 CMP R3, #0
00001014:0A000002 BEQ END
00001018:E1530001 CMP R3, R1
0000101C:02822001 ADDEQ R2, R2, #1
00001020:EAFFFFF9 B SCAN
00001024: END:
00001024:EF000011 SWI 0x11
00001028: S: .asciz "banana"
.byte 0
OutputView
Console Stdin/Stdout/Stderr
Execution ending, Instruction Count:43 Elasped Time:00:00:00.0103142
Instructions per second:4169
OutputView WatchView